



H33/H33-10 – Propeller Velocity Flowmeter

Used to measure very low point-velocities in water and other conductive fluids, this flowmeter uses the impedance of a rotating multi-bladed propeller to indicate rotational speed caused by the flowing fluid.

PRODUCT CODE: H33 / H33-10

Categories: Educational, H Series - Hydraulic Instruments, H33/H33-10 - Propeller Velocity Flowmeter

Tags: 5-Bladed Propeller , Battery Powered , Data Logger , Digital Display , Digital Indicator , Field Applications , Fully Portable , H Series , H33 , H33-10 , Hydraulic Measurement Instruments , Indicate Rotational Speed , Jewelled Bearing , Laboratory Applications , Mains Powered , Miniature Propeller Probes , Propeller Velocity Flowmeter , Sensing Head , Water Velocity

DESCRIPTION

H33 – PROPELLER VELOCITY FLOWMETER

Used to measure very low point-velocities in water and other conductive fluids, this flowmeter uses the impedance of a rotating multi-bladed propeller to indicate rotational speed caused by the flowing fluid.

The fine diameter of the sensing head enables the meter to be used in small ducts and channels with an ability to measure fluid velocity as low as 25 mm/sec.

A probe and digital indicator are required in combination to perform measurements but the digital indicator is supplied separately from the probes to allow an appropriate probe to be selected to suit specific applications.

Miniature Propeller Probes (H33-1/H33-2/H33-3)

Each probe consists of a sensing head that is mounted at the end of a slim stainless steel stem. A co-axial connector at the top of the stem allows the probe to be connected to the H33-10 Digital Indicator.

The sensing head consists of a 5-bladed propeller that runs in jewelled bearings. The propeller is able to rotate freely within a protective shrouded cage. An insulated gold electrode terminates 0.1mm from the tip of the rotating blades and detects the change in electrical impedance as each blade passes. As there is no physical contact between the electrode and blade, the propeller can rotate freely and respond to low velocity of the surrounding fluid.

H33-10 – DIGITAL INDICATOR

The digital indicator has been developed for use with the miniature propeller probes H33-1/2/3 where laboratory or field measurement of water velocity is required.

The digital indicator may be battery powered or mains powered using the standard power adaptor supplied. The power adaptor is intended for world-wide use and has four interchangeable plugs to suit all regions. The unit can operate on 110 Volts or 230 Volts at 50Hz or 60Hz.

When connected to an appropriate probe, the unit will indicate frequency of the pulses from the propeller that can be converted to water velocity using the calibration chart supplied with the probe. The unit can be programmed to read water velocity directly in units of cm/sec by entering a calibration constant for the probe in use (obtained from the chart supplied with each probe). Alternatively the unit can be programmed to count the pulses continuously, allowing averaging of the velocity measurement over a longer (user defined) period.

A 0-5 V DC output allows the unit to be connected to a suitable data logger to provide a permanent record of the variations in water velocity with time. A co-axial lead, 3 metres long, is supplied with the digital indicator for connecting it to any of the miniature propeller probes described above.

TECHNICAL SPECIFICATIONS

TECHNICAL DETAILS H33

Rotor: 11.6 mm diameter, machined plastic (balanced)

Spindle: Hardened stainless steel with conical ends

Bearings: Synthetic sapphire vee jewels

Cage: Nickel plated bronze

Stem: Stainless steel

Electrical connector: Co-axial

Weight: 0.20kg

Immersion length: 420mm maximum

TECHNICAL DETAILS H33-10

Power: Nickel metal hydride battery or mains power

Battery life: Typically 300 hrs on full charge

Display: Backlit multifunction LCD display 70 x 38 mm

Controls: On/off switch, and A + B buttons, Backlight button

Input: BNC miniature socket for co-axial cable

Output: 0.5 V DC – 100 MS update rate from 3.5 mm socket

Case size: 210 x 100 x 32 mm overall Moulded ABS plastic

Weight: 410g with batteries fitted

FEATURES & BENEFITS

The use of two probes allows the range of detectable velocities to be extended up to 300 cm/sec.

The indicator is supplied, as standard, with a set of rechargeable batteries. It can also be mains powered via the supplied universal charger.

All components have been chosen carefully to give a long reliable life with minimal changes in calibration within the operating ranges stated under Technical Data.

Capabilities

- Measurement of velocity in clean conductive fluid within the range 25 to 1500mm/sec (up to 3000mm/sec with the high speed probe)
- Operation in confined spaces with limited intrusive effects
- Suitable for both laboratory and field applications – battery powered (rechargeable) and fully portable
- Measurements are indicated on the digital display and may be connected to a data logger for later analysis

Accessories

H1-11: Adjustable Tripod stand with mountings



Shipping Specifications

H33/1/2/3

- **Volume:** 0.1m³
- **Gross Weight:** 2Kg

H33-10

- **Volume:** 0.1m³
- **Gross Weight:** 5Kg

Order Codes

- H33
- H33-1
- H33-2
- H33-3
- H33-10

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H1-11 – TRIPOD STAND

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